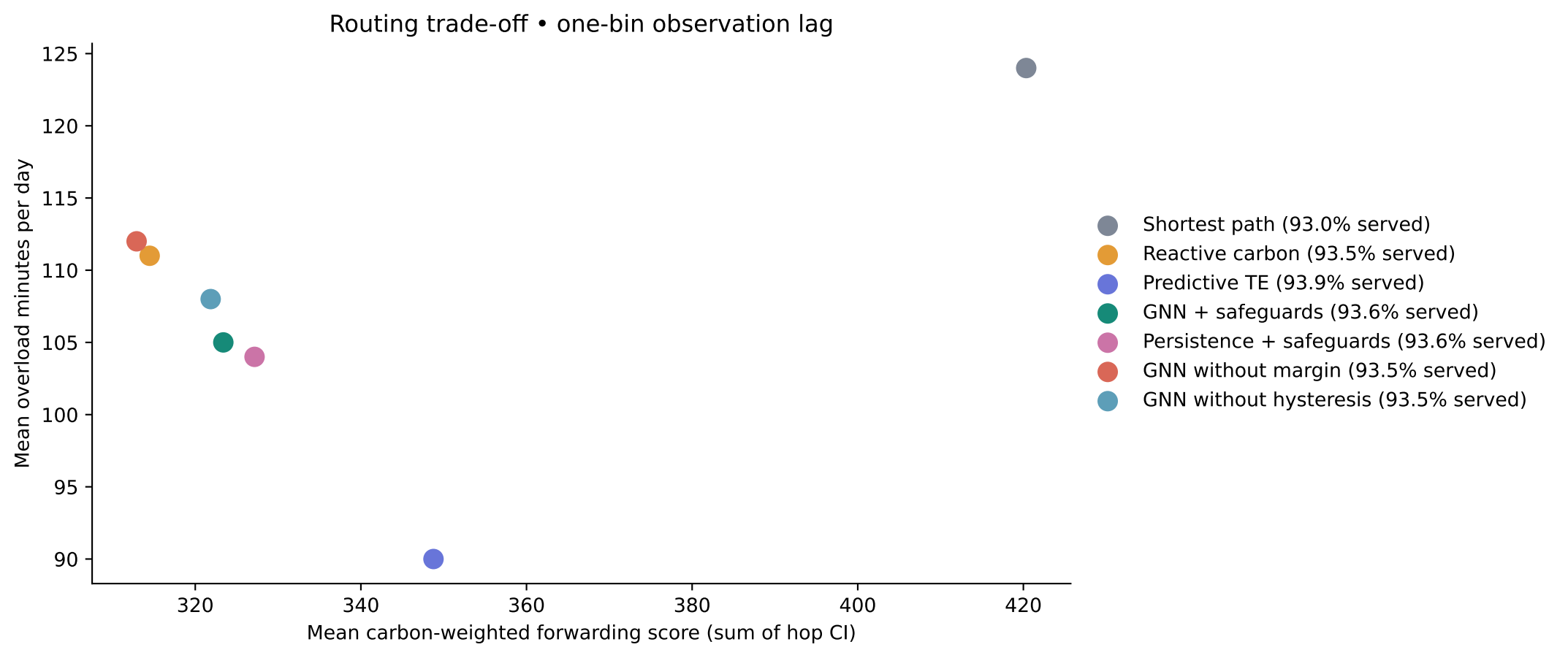
Routing trade-off • one-bin observation lag
Mean overload minutes per day
125
120
115
110
105
100
95
90
320
340
360
380
400
420
Mean carbon-weighted forwarding score (sum of hop CI)
Shortest path (93.0% served)
Reactive carbon (93.5% served)
Predictive TE (93.9% served)
GNN + safeguards (93.6% served)
Persistence + safeguards (93.6% served)
GNN without margin (93.5% served)
GNN without hysteresis (93.5% served)